CS-540 Data Literacy & Visualizations

2-3 Submit Activity: Clean It, Code It, Frame It

Malgorzata Debska

Southern New Hampshire University

Prof. Mahmoud Khasawneh

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Reflection: Transforming Raw Data into a Tabular Format

Transforming raw data into a structured, tabular format is a critical step in supporting exact analysis and effective decision-making in the field of business analytics. Real-world data is rarely clean or ready for immediate use. It often has missing values, inconsistent formatting, duplicate records, or unclear column headers. Without addressing these issues, any analysis performed on the data may produce misleading or unreliable results. Cleaning and structuring data ensures that insights are based on correct, consistent information.

Organizing raw data into a table improves clarity and usability by clearly defining variables and observations. In a tabular format, each column stands for a specific attribute, and each row is a single record, making patterns and relationships easier to show. This structure allows analysts to efficiently sort, filter, and summarize data, which is essential for tasks such as trend analysis, performance measurement, and forecasting. In business settings, well-structured data supports faster and more confident decision making by stakeholders who rely on correct reports and visualizations.

Data cleaning also plays a significant role in keeping data integrity. Standardizing formats, correcting data types, and handling missing values reduces the risk of errors during analysis. For example, ensuring numerical values are stored as numeric data types allows for right calculations, while consistent column names improve readability and reduce confusion when collaborating with others. These practices are especially important in professional environments where data is shared across teams and used repeatedly for different analytical purposes.

Ultimately, transforming raw data into a clean, tabular format creates a reliable foundation for deeper analysis and communication. In business analytics, decisions related to strategy, operations, and customer behavior depend on trustworthy data. Preparing data correctly at the beginning of the analysis process, analysts can produce meaningful insights, support evidence-based decisions, and communicate findings clearly to both technical and non-technical audiences.